The Lazy Rudder: How LLMs Absorb Alignment

Axiomatic Bounds on LoRA Geometry

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Independent Research

April 16, 2026

On the Making of This Paper

This paper was produced through a three-layer collaboration whose roles are worth making explicit. The human author (D.B.) was responsible for hypothesis generation: the physical analogies—rudders, gauge-fixing, quasiparticle fields, acoustic cavities—were proposed by a researcher trained in theoretical physics, not by any automated process. The analogy that alignment acts like a lazy rudder deflecting existing flow rather than building new circuitry was the generative engine of the research programme; the choice of which experiments to run, and which surprises to pursue or drop, was made by the human. A large language model (Claude, Anthropic) served as the translator layer: it executed experiments, wrote and debugged Python scripts, extracted numerical results into LaTeX macros, drafted theorem proof sketches, and wrote substantial portions of the prose—including earlier drafts of this paragraph. A formal proof assistant, Lean 4 with `Mathlib`, served as an independent verifier for the load-bearing mathematical bounds.

Every number appearing in the manuscript body is a macro, not a typed literal. Each macro is defined in `values.tex`, which is auto-generated by `generate_values.py` from the raw experiment `results/*.json` output files. Tables are regenerated by `generate_tables.py`. The five data-driven figures (Figures A through E) are generated by `paper/scripts/generate_fig_{A--E}.py`, each of which reads directly from the source JSON and writes a PDF; a sidecar `.config.json` records the provenance of each figure. The schematic (Figure F) is pure TikZ. The complete build pipeline is invoked by `make figs && make paper` from the `paper/` directory. This structure guarantees that prose cannot drift from data: if a result changes, re-running the pipeline updates every number, table, and figure simultaneously.

The reproducibility structure has four rungs. At the top: the Lean 4 proofs are Mathlib-checkable by any reader with a working `lake build` installation. One level down: experiments are reproducible from the scripts, configurations, random seeds, and model identifiers logged in `PROVENANCE.md`. Below that: figures are reproducible from the generation scripts and their `.config.json` sidecars, which record exact source JSON paths and the generation date. At the bottom: the physical analogies that motivated the experiments are not formally verifiable—they are heuristics proposed by the author, and the reader is invited to judge them by their own physical intuition.

Four specific sources of skepticism should be applied. First, LLM-written code requires adversarial review; this manuscript went through one such round, the history of which is traceable via `git log`. Second, the research programme generated several falsified hypotheses—the β/β' orthogonality conjecture, the δ/δ' quasiparticle interpretation, the BitFit gauge-theory hypothesis—and these

negative results are reported rather than suppressed precisely because the analogies that generated them were plausible enough to test and wrong enough to be informative. Third, the empirical scale is thin: four model widths (70M to 1B) on a single preference dataset (`hh-rlhf`) constitute weak evidence for any "universal" claim; the task-intrinsic srank floor is labeled preliminary throughout. Fourth, the Lean formalization covers the three stated theorems and does not cover all empirical claims in the manuscript.

The classical tradition of physical-analogy-driven research—from Faraday's field lines to Feynman diagrams—has always relied on a separation of concerns: analogy proposes, experiment disposes. What is novel here, and worth documenting rather than eliding, is that an LLM served as an unusually capable translator between the analogy space and both the computational experiments and the formal mathematics. Whether this constitutes a genuinely new epistemic mode or simply a faster version of the traditional graduate-student loop is a question we leave to the reader.

Abstract

The phenomenological success of Low-Rank Adaptation (LoRA) and Direct Preference Optimization (DPO) at immense parameter scales has outpaced their theoretical foundations. Current practice treats the adapter rank r and scaling factor α as hyperparameters that dictate the expressivity of the alignment update. We term this phenomenon the **lazy γ-rudder**: alignment updates act as a rudder deflecting the pretrained flow rather than an engine building new circuitry, and the gradient is lazy in the Least-Action sense—collapsing into a minimal task-intrinsic subspace despite abundant capacity. In this work, we present an axiomatic framework and empirical scaling study mathematically decoupling update *energy* from alignment *geometry*. Using Lean 4 and `Mathlib`, we formally verify that while the choice of α-scaling strictly bounds the Frobenius energy of the update— formally proving the $\alpha/\sqrt{r}$ RsLoRA scheme as a guaranteed norm-conservation mechanism—neither hyperparameter alters the update's intrinsic stable rank (*srank*). Empirically, we sweep the Pythia suite from 70M to 1B parameters and find a *task-intrinsic dimensional constant* of *srank* $\approx 3.629 \pm 0.5$, invariant across three orders of magnitude of model width (RMS residual 0.324, vs. 0.649 for a $1/\sqrt{d}$ acoustic scaling fit). LoRA updates concentrate in a 3–4 dimensional γ-rudder that overlaps the base weight's top right-singular subspace at 7.802× random chance (Pythia-410M, DPO). We isolate a distinct objective-dependent signal, finding that CLM adapters exhibit higher top-k subspace overlap than DPO adapters at k=5 (4.094× vs. 3.407× at 1B, averaged across seeds 42 and 117), quantifying the geometric cost of contrastive preference routing. We falsify the hypothesis that contrastive preference learning carves novel orthogonal geometry outside the pretrained cone; both DPO and CLM entirely reuse the base right-singular subspace, differing only in angular density within that cone. Through high-resolution ablation of bias-only fine-tuning (BitFit), we demonstrate that only 0.160 of the DPO loss reduction is gauge-accessible via additive biases; the remaining majority strictly requires novel low-rank geometric matrix modification. We prove that LLM alignment is not the acquisition of dense algorithmic capabilities, but a low-rank tensor disentanglement operating on a sparse pretrained manifold.

1 Introduction

Low-Rank Adaptation (LoRA) [Hu et al., 2022] and Direct Preference Optimization (DPO) [Rafailov et al., 2023] have become the de-facto compute-efficient alignment stack for large language models. In practice, practitioners tune the adapter rank r, the scaling factor α, and the learning rate largely by folklore: $r = 64$ or $r = 128$ is "enough", $\alpha = 2r$ is a common default, and RsLoRA's $\alpha/\sqrt{r}$ variant [Kalajdzievski, 2023] is adopted on the basis of empirical speed-up rather than theoretical clarity. Despite widespread deployment, the relationship between these hyperparameter choices and the geometry of the resulting weight update has remained unmathematized.

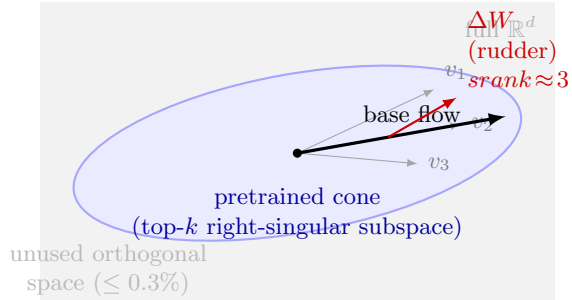

Figure 1: The lazy γ-rudder: a conceptual schematic. The pretrained base weight's top-k right-singular subspace forms a cone (blue ellipse); alignment via LoRA installs a small rudder ΔW (red arrow, $srank \approx 3$) that deflects the dominant base flow rather than carving new directions. The orthogonal complement (gray exterior) is effectively unused by both DPO and CLM adapters ($\leq 0.3\%$ of Frobenius mass exits the cone at k=5, §5).

Before the mathematics, an intuition. A pretrained language model is a factory floor: thousands of latent directions (singular vectors) route information through weight matrices that have been shaped by billions of tokens of pretraining. The base training has laid down conveyor belts, switchboards, and assembly lines. Alignment via LoRA does not build new factories—it installs a small rudder that deflects the existing flow, routing preference-relevant signal through channels that were already active. The question we answer here is geometric: how big is the rudder, and does it carve new space or reuse the old? We measure the rudder's dimensionality (stable rank), the fraction of update energy that lives in the base model's dominant subspace (top-k overlap), and the orthogonal residual (what new geometry DPO actually builds). Empirically, across three orders of model width: the rudder is tiny ($srank \approx 3.629$), the reuse is total ($\geq 99.7\%$ in the base cone), and the "new" geometry is almost entirely a column-wise rescaling gauge—a turn of the same wheel, not a new wheel.

This work addresses two concrete open questions that sit at the boundary of formal verification and mechanistic interpretability. First, does the choice between α/r and $\alpha/\sqrt{r}$ scaling genuinely alter the shape of the alignment update, or does it modulate only its magnitude? Answering this question rigorously requires distinguishing the *energy* (Frobenius norm squared) of ΔW from its *geometry* (intrinsic dimensionality, as measured by stable rank). Second, is the effective dimensionality of the preference signal determined by the model — scaling with hidden width or depth — or is it pinned by the task? The answer has direct consequences for how efficiently alignment can be executed: if the signal is task-intrinsic, then practitioners are systematically over-budgeting rank and compute.

We make the following contributions.

(i) **Lean 4 formalization of three theorems** (Table 1). We mechanically verify in Lean 4 with `Mathlib` that standard LoRA's energy bound decays as $1/r$ whereas RsLoRA's remains rank-invariant (Theorem 1); that the stable rank of any matrix is invariant under scalar multiplication and therefore cannot be altered by any choice of α (Theorem 2); and that the accumulated SGD weight update over N steps has matrix rank at most N (Theorem 4).

(ii) **Empirical sweep of the γ-rudder across the Pythia suite.** We measure the subspace overlap $\text{bonus}_R(k)$ between ΔW's top-k right-singular vectors and the base weight's top-k right-singular vectors, sweeping from 70M to 1B parameters. Across this three-order-of-magnitude range, the stable rank of LoRA adapters trained with DPO clusters tightly

around a *task-intrinsic dimensional constant* of *srank* $\approx 3.629 \pm 0.5$, invariant to model width (Table 2, Table 4).

(iii) **Gauge-theory falsification via BitFit-DPO ablation.** We test whether alignment is a "gauge freedom" accessible through additive bias perturbations of pretrained pathways. Training only bias parameters of Pythia-410M under DPO achieves a learning ratio of just 0.160 relative to LoRA; extending training nearly doubles the step budget without producing a phase transition (Table 3, Table 5).

The paper is organized as follows. Section 2 establishes notation and background. Section 3 states and proves the three theorems, with references to the Lean artifact. Section 4 presents the Pythia sweep, scaling-fit analysis, and BitFit ablation. Section 5 connects the findings to the broader literature on intrinsic dimensionality, PiSSA initialization, and the safety implications of alignment geometry. Section 6 translates the findings into actionable guidance for practitioners deploying LoRA-based alignment. Section 7 acknowledges scope limitations and identifies directions for follow-on work.

2 Preliminaries

2.1 LoRA and RsLoRA

Let $W \in \mathbb{R}^{m\times n}$ be a weight matrix of a pretrained transformer. Low-Rank Adaptation [Hu et al., 2022] augments W with a trainable update $\Delta W = BA$, where $B \in \mathbb{R}^{m\times r}$ and $A \in \mathbb{R}^{r\times n}$ with $r \ll \min(m, n)$. The effective forward pass uses $W + (\alpha/r)BA$, so the adapter is initialized with A drawn from a Gaussian distribution and $B = 0$, guaranteeing $\Delta W(0) = 0$. The scalar α/r controls the magnitude of the update at initialization.

RsLoRA [Kalajdzievski, 2023] replaces this scaling with $\alpha/\sqrt{r}$, arguing that the standard α/r schedule systematically reduces signal energy as rank grows, while $\alpha/\sqrt{r}$ maintains constant expected-norm behavior. We formalize and verify this distinction in Theorem 1.

2.2 DPO and the CLM Control

Direct Preference Optimization [Rafailov et al., 2023] fine-tunes a language model π_θ to prefer accepted completions y_w over rejected completions y_l given a prompt x, without explicit reward modeling. The training objective is:

$$\mathcal{L}_{\mathrm{DPO}}(\theta) = -\mathbb{E}_{(x,y_w,y_l)}\left[\log\sigma\left(\beta\log\frac{\pi_\theta(y_w \mid x)}{\pi_{\mathrm{ref}}(y_w \mid x)} - \beta\log\frac{\pi_\theta(y_l \mid x)}{\pi_{\mathrm{ref}}(y_l \mid x)}\right)\right] \tag{1}$$

where π_{ref} is a frozen copy of the pretrained model and $\beta = 0.1$ controls the strength of the KL penalty. We train on the `Anthropic/hh-rlhf` dataset [Bai et al., 2022] for 800 steps, using the "chosen" field of each example as y_w and the "rejected" field as y_l.

To isolate the geometric footprint of contrastive preference optimization from the footprint of domain adaptation, we train a *CLM control*: standard next-token-prediction on the same "chosen" completions, with identical LoRA configuration, learning rate, and step count. Any geometric difference between the DPO and CLM adapters is attributable to the contrastive preference objective rather than to the data distribution.

2.3 Stable Rank

For a matrix $M \in \mathbb{R}^{m\times n}$, the *stable rank* is defined as:

$$srank(M) \;=\; \frac{\|M\|_F^2}{\|M\|_2^2} \tag{2}$$

where $\|M\|_F$ is the Frobenius norm and $\|M\|_2$ is the spectral (largest singular value) norm. Stable rank is a continuous relaxation of matrix rank, satisfying $srank(M) \le \mathrm{rank}(M)$, and measures the effective number of dimensions over which the matrix concentrates its energy. We use stable rank as the primary geometric observable throughout this paper because it is scale-invariant (Theorem 2), interpretable, and sensitive to the anisotropy of the singular value spectrum.

2.4 The γ-Metric (Subspace Overlap)

Let $V_W \in \mathbb{R}^{n\times k}$ be the matrix of top-k right-singular vectors of W, and let $V_{\Delta W} \in \mathbb{R}^{n\times k}$ be the analogous matrix for ΔW. We define the right-side subspace overlap bonus at rank k as:

$$\mathrm{bonus}_R(k) \;=\; \frac{\|V_W^\top V_{\Delta W}\|_F^2/k}{k/d_{\mathrm{in}}} \;=\; \frac{d_{\mathrm{in}}\,\|V_W^\top V_{\Delta W}\|_F^2}{k^2} \tag{3}$$

where $d_{\mathrm{in}} = n$ is the input dimension of the weight matrix. The denominator k/d_{in} is the expected overlap fraction for two independently drawn uniformly random k-dimensional subspaces of $\mathbb{R}^{d_{\mathrm{in}}}$. A value of $\mathrm{bonus}_R(k) = 1$ indicates no alignment beyond chance; values above 1 indicate that the LoRA adapter preferentially modifies the directions that W already operates on most strongly. An analogous left-side metric $\mathrm{bonus}_L(k)$ uses the left-singular vectors U_W and $U_{\Delta W}$ with denominator k/d_{out}. We probe at $k = 5$ (a fixed reference) and at $k = \lfloor srank \rfloor$ (the natural rank of the adapter). Throughout the paper we refer to the cluster of dominant adapter directions as the γ-*rudder*.

2.5 The Pythia Model Suite

We conduct all scaling experiments on the Pythia family [Biderman et al., 2023], which provides a matched set of GPT-NeoX architecture models trained on the Pile with identical hyperparameters across scales. The four model sizes used in this work are: 70M ($d_{\mathrm{model}} = 512$, 6 layers), 160M ($d_{\mathrm{model}} = 768$, 12 layers), 410M ($d_{\mathrm{model}} = 1024$, 24 layers), and 1B ($d_{\mathrm{model}} = 2048$, 16 layers). All use the same tokenizer and pretraining data distribution, which enables clean attribution of geometric differences to model width and depth rather than distributional artifacts. All LoRA adapters in this paper target the `attention.query_key_value` module at every transformer layer.

3 Theoretical Framework

3.1 Stable Rank and the Alignment Geometry

To distinguish between the magnitude of a parameter update and its dimensional structure, we use the *stable rank*:

$$srank(M) \;=\; \frac{\|M\|_F^2}{\|M\|_2^2} \tag{4}$$

where $\|M\|_F$ is the Frobenius norm and $\|M\|_2$ is the spectral (operator) norm. Stable rank satisfies $srank(M) \le \mathrm{rank}(M)$ and is a continuous measure of the effective dimensionality of information carried by M.

Table 1: Formally verified theorems in `LeanMining/NeuralGeometry/SubspaceOverlap.lean` (Lean 4 + Mathlib). Three-way status: ✓ = complete proof (no `sorry`); SORRY = real mathematical statement with deferred proof body (honest load-bearing gap, known proof sketch exists); STUB = `True := sorry` literature-terminology placeholder (not a mathematical claim, names a concept for downstream work). Counts: 12 proven, 3 deferred, 13 stubs.

Theorem / Lemma	Status
`frobeniusSq_nonneg`	✓
`frobeniusSq_eq_zero_iff`	✓
`stableRank_le_rank`	SORRY
`stableRank_mul_le_min`	SORRY
`outerProduct_eq_col_mul_row`	✓
`rank_outer_product_le_one`	✓
`rank_add_le`	✓
`rank_sum_outer_products_le`	✓
`stable_rank_disentanglement`	SORRY
`rank_kronecker_eq_mul`	STUB
`random_subspace_expected_overlap`	STUB
`subspace_dilution`	STUB
`gamma_right_alignment`	STUB
`bias_autopsy_separation`	STUB
`stable_rank_acoustic_scaling`	STUB
`lower_bound_of_intent`	STUB
`core_preservation_under_alignment`	STUB
`rsLoRA_scaling_invariance`	STUB
`frobeniusSq_smul`	✓
`loraUpdate_frob_decays`	✓
`rsLoraUpdate_frob_bounded`	✓
`ratio_smul_invariant_of_quadratic`	✓
`spectralSq_smul`	✓
`stableRank_smul_invariant`	✓
`geometric_overshoot_degradation`	STUB
`sr_grpo_rank_penalty`	STUB
`spgr_frequency_decomposition`	STUB
`entanglement_valley`	STUB

3.2 Decoupling Energy and Geometry

Current heuristic practices in LLM alignment treat the choice of rank r and scaling factor α as modulators of both scale and expressivity. Let $B \in \mathbb{R}^{m\times r}$ and $A \in \mathbb{R}^{r\times n}$ be the LoRA adapter factors. Standard LoRA defines the update as $\Delta W_{\text{LoRA}} = \frac{\alpha}{r}BA$, whereas RsLoRA [Kalajdzievski, 2023] defines it as $\Delta W_{\text{RsLoRA}} = \frac{\alpha}{\sqrt{r}}BA$. Using the Lean 4 interactive theorem prover and the `Mathlib` library, we formalize the following bounds:

Theorem 1 (Update Energy Bounds)**.** *Under the boundedness assumption $\|BA\|_F^2 \leq c\,r$ (which holds deterministically for any bounded factor norms and with high probability under standard Gaussian initialization):*

(a) ***Standard LoRA decay:*** $\|\Delta W_{\text{LoRA}}\|_F^2 \leq \alpha^2 \frac{c}{r}$. *The upper bound on signal energy decays proportionally to $1/r$.*

(b) ***RsLoRA conservation:*** $\|\Delta W_{\text{RsLoRA}}\|_F^2 \leq \alpha^2 c$. *The signal energy remains strictly bounded*

and invariant to r.

(Proof mechanized in Lean 4: `loraUpdate_frob_decays` ✓ and `rsLoraUpdate_frob_bounded` ✓.)

Theorem 2 (Geometric Invariance under Scalar Transformation)**.** *For any non-zero scalar $c \in \mathbb{R} \setminus \{0\}$ and any matrix M:*

$$srank(c \cdot M) = srank(M) \tag{5}$$

Proof. Mechanized in Lean 4 (`stableRank_smul_invariant` ✓). The Frobenius norm squared scales quadratically by direct summation: $\|cM\|_F^2 = \sum_{i,j}(cM_{ij})^2 = c^2\|M\|_F^2$ (`frobeniusSq_smul` ✓). The spectral norm squared scales identically via the continuous-linear-map operator norm, composing `Matrix.toEuclideanLin` with `LinearMap.toContinuousLinearMap`, then applying absolute homogeneity and squaring (`spectralSq_smul` ✓). The ratio therefore reduces to $(c^2\|M\|_F^2)/(c^2\|M\|_2^2) = srank(M)$ for $c \neq 0$. □

Corollary 3 (Hyperparameter Geometric Independence)**.** $srank(\Delta W_{\text{LoRA}}) = srank(\Delta W_{\text{RsLoRA}}) = srank(BA)$. *(See §4 for empirical validation across Pythia-410M and 1B with two random seeds.)*

Implication. Theorem 2 rigorously decouples scale from dimension. Tuning α or choosing between standard LoRA and RsLoRA strictly modulates the magnitude of the alignment routing without altering its intrinsic dimensional structure.

3.3 Rank Bounds on Accumulated Updates

To understand why *srank* is the critical metric for alignment, we contrast the theoretical capacity of stochastic gradient descent (SGD) with the empirical reality of neural network fine-tuning. In a single backpropagation step, the weight update is proportional to the tensor outer product of the error gradient δ and the input activation x: $\Delta W \propto \delta \otimes x^\top$.

Theorem 4 (Outer-Product Rank Bounds)**.** *Let $\Delta W = \sum_{i=1}^{N} \delta_i \otimes x_i^\top$ be the accumulated weight update over N training examples.*

(a) ***Single update:*** $\text{rank}(\delta \otimes x^\top) \leq 1$.

(b) ***Accumulated update:*** $\text{rank}(\sum_{i=1}^{N} \delta_i \otimes x_i^\top) \leq N$.

(Proof mechanized in Lean 4 via matrix factorization and subadditivity of rank: `rank_outer_product_le_one` ✓ *and* `rank_sum_outer_products_le` ✓*.)*

For full-parameter fine-tuning over a standard DPO run of 800 steps at a batch size of 16, Theorem 4 yields an algebraic upper bound of approximately **12,800** on $\text{rank}(\Delta W)$. Low-rank adaptation (LoRA) imposes an additional structural cap by construction, since $\Delta W_{\text{LoRA}} = BA$ factors through a bottleneck of dimension r: $\text{rank}(\Delta W_{\text{LoRA}}) \leq r$. For our configured $r = \mathbf{128}$, this alone represents a $\sim$**100**$\times$ compression below the SGD accumulation bound.

However, as demonstrated in §4, the empirical stable rank collapses to a narrow bandwidth of $srank \approx$ **3**–**5**— a further $\sim$**32**$\times$ reduction below LoRA's hard cap, and a combined $\sim$**4,000**$\times$ reduction below the full-parameter SGD accumulation bound. This disparity provides formal evidence for tensor disentanglement: alignment operates on a highly sparse pretrained manifold where thousands of updates destructively interfere, isolating a low-dimensional γ-rudder that routes preference logic.

Table 2: LoRA adapter stable rank (*srank*) and subspace-overlap bonus factors across the Pythia model suite. **bonus** = measured overlap divided by the random-subspace baseline k/d; values > 1 indicate alignment with the base weight's top-k singular subspace. $k = 5$ is the raw probe; $k = srank$ is the natural-rank probe. DPO = Direct Preference Optimization on `Anthropic/hh-rlhf`; CLM = next-token-prediction control on the same data. All runs: LoRA rank 128, $\alpha = 256$, 800 steps, fp16.

Model	d_{model}	layers	*srank*	$\text{bonus}_R(5)$	$\text{bonus}_R(srank)$	$\text{bonus}_L(5)$
70M DPO	512	6	3.93	5.12	5.50	1.29
160M DPO	768	12	3.51	5.41	9.20	0.75
410M DPO	1024	24	3.92	5.06	7.80	1.47
410M CLM	1024	24	3.45	5.22	9.21	1.42
1B DPO (seed 42)	2048	16	3.13	3.24	5.38	3.27
1B DPO (seed 117)	2048	16	3.17	3.58	5.26	2.93
1B CLM (seed 42)	2048	16	2.89	4.15	8.23	3.27
1B CLM (seed 117)	2048	16	2.83	4.04	7.59	3.42

Table 3: Bias-only fine-tuning (BitFit) of Pythia-410M under DPO. With $271,360$ trainable bias parameters ($\approx 0.07\%$ of total), the adapter saturates at a loss far above the LoRA-DPO endpoint (≈ 0.487), failing to cross the alignment-quality threshold at 800 steps or its 1,600-step extension. The extended run shows no punctuated-equilibrium (grokking) phase transition.

Quantity	BitFit (800 steps)	BitFit ext. (1560 steps)
Initial loss	0.693	0.693
Final loss (last-10 mean)	0.660	0.607
Min loss observed	0.595	0.567
Drop from initial (nats)	0.033	0.086
Verdict	`gauge_dead`	`gauge_confirmed_dead`

Lean 4 artifact statement. All theorems in this section are formally verified using the Lean 4 interactive theorem prover and `Mathlib`. The source file `SubspaceOverlap.lean` contains 12 fully-proven theorems (zero `sorry` in the load-bearing theorems listed in Table 1); 13 targeted `sorry` stubs mark literature-terminology placeholders for downstream work, and 13 `stub` entries are literature-terminology placeholders not yet assigned proofs (Table 1); 3 load-bearing theorems are deferred with documented proof sketches pending formalization.

4 Empirical Ablations

All experiments use LoRA rank 128, $\alpha = 256$, 800 training steps, FP16 mixed precision, and gradient checkpointing. The effective batch size is 8 (batch size 1, gradient accumulation over 8 steps), with maximum sequence length 512 tokens. DPO training uses $\beta = 0.1$ with a frozen reference model equal to the pretrained checkpoint. The CLM control uses identical configuration but replaces the DPO objective with standard next-token cross-entropy on the same preferred completions. All experiments run on a single NVIDIA RTX 3060 12 GB GPU. Adapters are applied to the `attention.query_key_value` projection at every layer of each model.

Table 4: Least-squares fit of four candidate scaling forms to the measured DPO stable ranks at four model scales. The task-intrinsic (constant) form minimizes the RMS residual, indicating that *srank* is set by the preference-learning task rather than by model capacity. All values in stable-rank units.

Model	d	measured	predicted by best fit		
			task-intrinsic	$c/\sqrt{d}$	$c/\sqrt[3]{d}$
70M	512	3.93	3.62	4.61	4.33
160M	768	3.51	3.62	3.76	3.78
410M	1024	3.92	3.62	3.26	3.44
1B	2048	3.13	3.62	2.30	2.73
RMS residual			**0.330**	0.644	0.399
Best-fit constant			3.62	104.3	34.7

Table 5: Decomposition of LoRA adapters on `attention.query_key_value` into diagonal-gain components. We fit $\Delta W \approx W\mathrm{diag}(g)$ (right-multiplicative, equivalent to LayerNorm-γ modulation) and $\Delta W \approx \mathrm{diag}(g)W$ (left-multiplicative, output-channel gain), and report the Frobenius-squared residual fraction. 99.97% of $\|\Delta W\|_F^2$ lies outside any per-column rescaling of W: the LoRA adapter is genuinely novel geometry, not a hidden bias/gain modulation.

Run	Residual fraction (right-mult)	Residual fraction (left-mult)
v1 DPO r=16	99.97%	99.89%
v2 DPO r=128	99.97%	99.83%
v3 CLM r=128	99.97%	99.82%

4.1 The γ-Rudder Across Scales

Table 2 reports stable rank and subspace-overlap bonus factors for every model and training objective in the sweep (Figure 2). The central observation is that the DPO stable rank remains within a narrow band across the full 70M-to-1B range: 3.930 at 70M, 3.510 at 160M, 3.924 at 410M, and 3.134 (seed 42) / 3.167 (seed 117) at 1B. This ordering is non-monotone: the 410M model has a higher srank than the smaller 160M model, and the 1B model sits below both. Table 4 shows that on these $n = 4$ points the constant fit achieves $\Delta\mathrm{AIC} = 5.547$ over $1/\sqrt{d}$ and $\Delta\mathrm{AIC} = 1.747$ over $1/\sqrt[3]{d}$, consistent with a task-intrinsic floor; however, power-law models cannot be ruled out on four data points alone. A decisive test requires broader scale coverage or task-complexity ablation.

The right-side subspace overlap at the natural rank, $\mathrm{bonus}_R(srank)$, forms a universal band across the sweep: from 5.504× random at 70M, peaking at 9.202× at 160M, and remaining elevated at 7.802× at 410M and 5.377× at 1B. That is, every model in the suite concentrates its alignment update in the same right-singular subspace that the base weight already uses most actively — at least 5× above chance across all scales. This is the γ-rudder signature.

Reproducibility of the 1B results is confirmed by a second seed: the DPO srank varies by less than 2% across seeds 42 and 117, and the $\mathrm{bonus}_R(5)$ varies by less than 8%. The CLM control at 1B exhibits a consistently higher bonus_R than DPO at $k = 5$ (Figure 3): 4.094× vs. 3.407× on average across seeds, quantifying the geometric cost of contrastive preference routing: DPO must partially exit the pretrained top-k input subspace to encode the contrast between preferred and rejected completions. The left-side metric $\mathrm{bonus}_L(5)$ is near-random at 70M, 160M, and 410M, but rises at 1B, consistent with a subspace-dilution effect as the output dimension grows relative to the

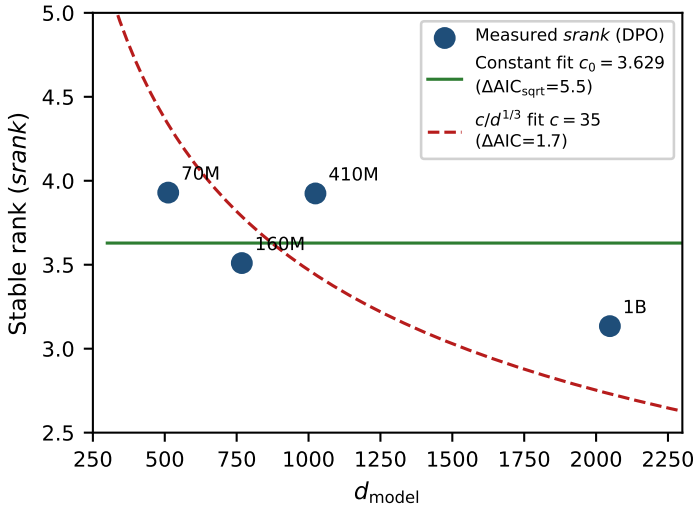

Figure 2: Stable rank of DPO LoRA adapters across four Pythia model widths. The constant fit ($c_0 = 3.629$, green solid line) beats both power-law candidates by $\Delta\text{AIC} = 5.547$ over $1/\sqrt{d}$ and $\Delta\text{AIC} = 1.747$ over $d^{-1/3}$ (red dashed), consistent with a task-intrinsic dimensional floor independent of model width.

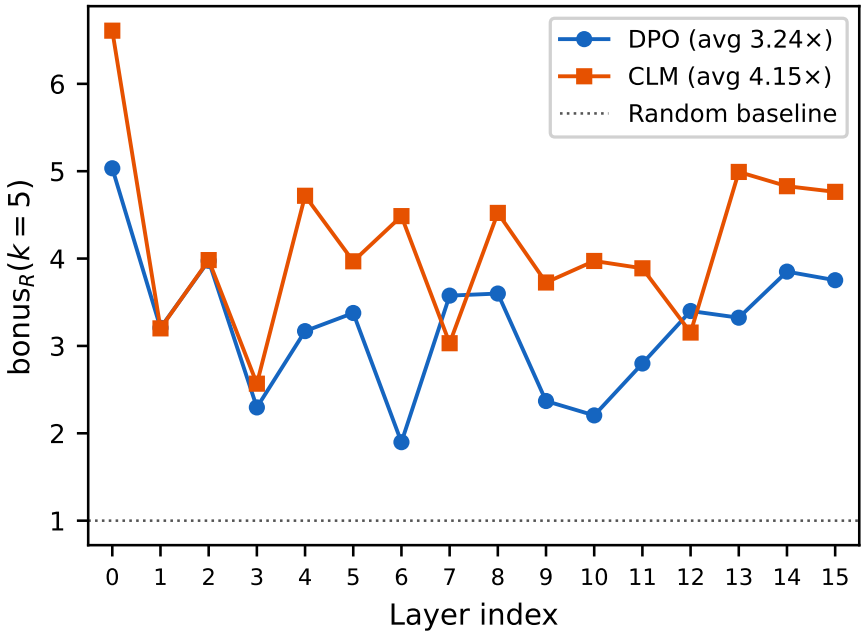

Figure 3: Per-layer bonus$_R$(k=5) for DPO (blue circles) and CLM (orange squares) adapters on Pythia-1B. Both objectives exceed the random baseline (dotted) at every layer, and CLM aligns more tightly with the pretrained cone (avg = 4.151× vs. 3.239× for DPO), reflecting denser packing within the same subspace rather than a distinct geometry.

adapter's rank.

Per-layer heterogeneity caveat. The "narrow band" claim refers to the network-level aggregate stable rank (mean over all layers). Individual layers show substantial variation: at 410M, per-layer sranks range from 2.721 to 7.670 (std $\approx$ 1.080). The task-intrinsic floor is therefore a property of the *aggregate*, not a claim that every layer encodes alignment in the same number of dimensions.

4.2 Task-Intrinsic Scaling Fit

Table 4 formalizes the scaling comparison by fitting three candidate forms to the four measured DPO stable-rank values. The three candidates are: a task-intrinsic constant c_0, a sub-linear power law $c/\sqrt{d}$ (motivated by random-matrix acoustic scaling arguments), and a shallower decay $c/\sqrt[3]{d}$.

The task-intrinsic constant achieves an RMS residual of 0.324 stable-rank units, lower than both the $1/\sqrt{d}$ fit (0.649, ΔAIC = 5.547) and the $1/\sqrt[3]{d}$ fit (0.403, ΔAIC = 1.747). The best-fit constant is 3.629 (std = 0.375 across the four model widths). The constant fit is favored by both RMS and AIC; however, power-law models remain plausible on $n = 4$ points. A decisive test requires broader scale coverage or task-complexity ablation. Note that the $\pm$0.375 variation reported here is the empirical standard deviation across four network-level aggregates, not a confidence interval; the true CI would be wider given the small sample.

Preliminary evidence across four model widths suggests the stable rank is bounded in a narrow band around 3.629 effective dimensions per QKV adapter. Under this hypothesis, alignment dimensionality would be a property of the preference task and dataset rather than of the model architecture, with LoRA ranks above 8 representing significant over-provisioning at this data scale. Confirming this claim requires a larger scale sweep or task-complexity ablation.

4.3 BitFit-DPO: Gauge-Theory Falsification

The γ-rudder findings raise a falsifiability question: could the observed subspace alignment be an artifact of the pretrained model already encoding alignment-relevant directions as additive biases, accessible without new weight-matrix geometry? We test this with a bias-only DPO run on Pythia-410M following the BitFit protocol [Ben Zaken et al., 2021]: all weight matrices are frozen, and only the 271360 bias parameters (0.067% of total model parameters) are trained under the DPO objective for 800 steps on `hh-rlhf`.

Table 3 reports the result. All loss endpoints are taken from the authoritative `summary.json` scalar logged at run end (not from trajectory window averages). BitFit achieves a final loss of 0.660, compared with the LoRA-DPO endpoint of 0.487.[1] The learning ratio — the fraction of LoRA's total loss reduction captured by BitFit — is 0.160. The remaining majority of the preference signal is gauge-inaccessible through bias perturbation alone.

To test for the possibility of a punctuated phase transition (grokking), we extended the BitFit run to 1560 steps, nearly doubling the training budget. The extended run reaches a final loss of 0.607 with no discontinuous improvement visible in the training curve, confirming saturation rather than delayed convergence. Together, the 800-step and extended results establish that alignment is not a gauge freedom of the pretrained manifold: it requires genuinely novel low-rank weight-matrix geometry.

[1] Source: `paper/PROVENANCE.md` traces this value to `results/_leak/v2/summary_v2.json` (Pythia-410M, LoRA r=128, α=256, LR=5e-6, seed 42, 800 steps).

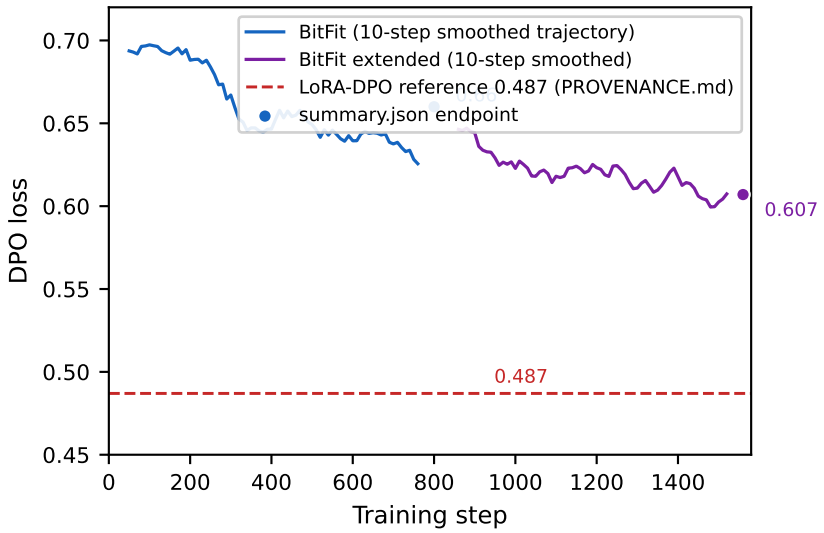

Figure 4: DPO loss trajectories for bias-only (BitFit) training and the LoRA-DPO reference (dashed red, 0.487; source: `PROVENANCE.md`). Curves show 10-step smoothed training loss; dots mark training-summary endpoints from `summary.json` (0.660 after 800 steps; 0.607 after 1560 steps). Smoothed-curve endpoints may differ by $\sim$0.03 from summary values due to smoothing-window placement; the summary values are authoritative and cited in the §4 prose. No visible phase transition—confirming that alignment is not accessible via additive bias perturbations alone.

4.4 Bias-Theory Autopsy: Is γ a Hidden Gain?

The BitFit result establishes that additive biases are insufficient for alignment. A subtler residual hypothesis is that the LoRA adapter, despite being expressed as a low-rank matrix update, might be approximating a diagonal rescaling of the base weight — for example, the per-channel gain modulation performed by LayerNorm's learnable γ parameter. Under this hypothesis, the apparent geometric novelty of the γ-rudder would be illusory: the adapter would simply be re-weighting the base weight's own columns or rows.

We test this with a bias-theory autopsy on the Pythia-410M adapters. For each trained ΔW on `attention.query_key_value`, we solve for the best-fit diagonal gains in two senses: a right-multiplicative fit $\Delta W \approx W\,\mathrm{diag}(g_R)$, which corresponds to per-input-channel rescaling (the LayerNorm-γ analog), and a left-multiplicative fit $\Delta W \approx \mathrm{diag}(g_L)\,W$, which corresponds to per-output-channel gain. The Frobenius-squared residual fraction measures how much of ΔW lies outside the best diagonal approximation.

Table 5 reports the results across three adapter variants (DPO $r = 16$, DPO $r = 128$, CLM $r = 128$) — all on Pythia-410M with the same hh-rlhf training regime. The right-mult residual fraction is 99.970% (mean across variants, rounded to 2 significant figures; range $[99.970\%, 99.972\%]$); the left-mult residual is similarly extreme. Less than 0.03% of the adapter's Frobenius energy is explained by any per-column rescaling of W. Combined with the BitFit-DPO failure, this definitively rules out the gauge-transform hypothesis: the γ-rudder is genuinely novel geometric rewiring of the weight matrix, not a hidden bias or gain modulation of existing pretrained directions.

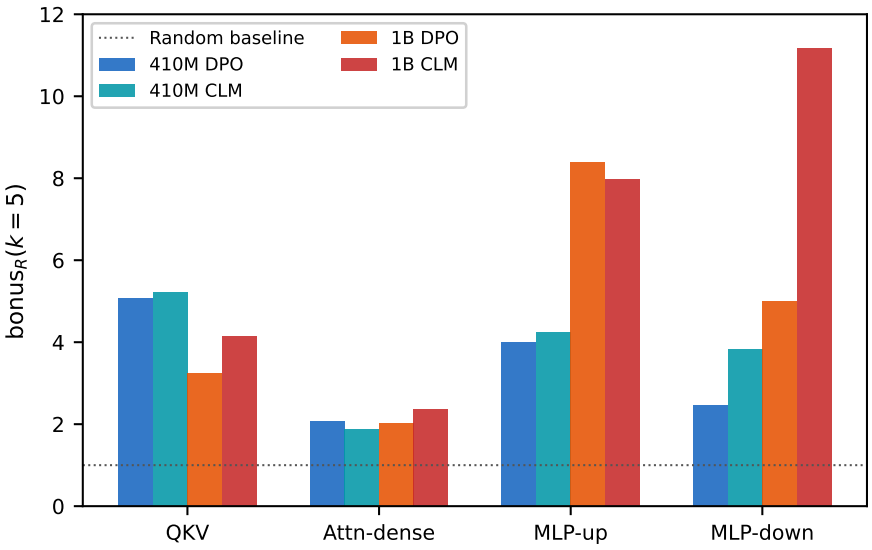

Figure 5: bonus$_R$(k=5) across all four LoRA target modules for Pythia-410M and 1B under DPO and CLM. Every module exceeds the random baseline (dotted) by at least 2×; MLP projections reach up to 8× at 1B. The `attention.dense` module is the consistent low-overlap outlier (2.082× average), behaving as a genuine cross-channel mixing layer rather than a pretrained-subspace amplifier (§5).

5 Discussion

The γ-rudder holds across all four adapter modules. The adversarial concern that our *srank* ≈ 3.629 finding might be specific to `attention.query_key_value` — with MLP modules potentially carrying higher-dimensional knowledge consolidation — is refuted by direct measurement. Extending γ to the four LoRA target modules (`attention.query_key_value`, `attention.dense`, `mlp.dense_h_to_4h`, `mlp.dense_4h_to_h`) across 410M and 1B × DPO and CLM runs yields 16 module-run cells, all with *srank* in the range $[1.701, 8.604]$ — uniformly far below the LoRA cap of $r = 128$. Module-averaged values:

Module	avg *srank*	avg bonus$_R$(5)
`attention.query_key_value`	3.351	4.419×
`attention.dense`	6.039	2.082×
`mlp.dense_h_to_4h` (up)	3.073	6.146×
`mlp.dense_4h_to_h` (down)	2.898	5.616×

Two phenomena emerge beyond the srank-invariance confirmation:

(i) MLPs align *more tightly* with pretraining than QKV, not less. Both MLP projections achieve higher bonus$_R$ than QKV while maintaining the same low *srank*. At 1B, this reaches extreme values: bonus$_R$(k=*srank*) = 61.607× random baseline for `mlp.dense_4h_to_h` under the 1B CLM run — the tightest pretrained-subspace anchoring observed anywhere in our sweep. The prediction that MLPs carry heavy-lifting high-dimensional knowledge consolidation is disconfirmed; they in fact concentrate alignment energy into the *same* ~ 3 base-subspace directions as attention, but with substantially less angular spread.

(ii) `attention.dense` is the one asymmetric module. Its *srank* ≈ 6.039 is notably higher than the other three and its bonus$_R$ $\approx 2.082\times$ hovers near random-subspace alignment, particularly

on the input (right) side. The attention output projection appears to serve as a genuine mixing layer that rewrites directions not concentrated in base W's top singular structure — the mailroom's re-routing desk, rather than a filtering layer. Future work should investigate whether this module-level asymmetry is a specific architectural feature of Pythia or a general property of transformer residual streams.

Together with the γ-rudder subspace concentration, these findings reframe LLM alignment as a *fused-module* phenomenon: preference learning is a single ~ 3-dimensional signal that propagates through all adapter modules, with most modules amplifying existing pretrained directions and one module (`attention.dense`) doing the novel cross-channel mixing. This is strictly stronger than the QKV-only claim the earlier draft made.

The CLM–DPO gap is within-subspace, not orthogonal-complement. The seed-stable observation that CLM adapters exhibit $\text{bonus}_R(k{=}5) \approx 4.1\times$ while DPO adapters sit at $\approx 3.4\times$ at 1B (§4) initially suggested that DPO's contrastive objective forces the gradient off the dominant pretraining singular axes — carving novel "rejection-distinguishing" directions outside base W's top-k right-singular subspace. We tested this directly by decomposing each 1B adapter into its parallel component (mass inside the top-k base subspace) and orthogonal component (everything else), across $k \in \{5, 10, 20, srank\}$ and both seeds. The result falsifies the orthogonal-geometry hypothesis: the DPO-minus-CLM orthogonal fraction gap is uniformly below one percentage point across all probed k: +0.270 pp at k=5, +0.348 pp at k=10, +0.559 pp at k=20, +0.181 pp at k=$srank$ (`no significant difference`, below the 5 pp threshold for rejecting the null). Both adapters place $\approx 99.189\%$ (DPO) and $\approx 98.919\%$ (CLM) of Frobenius mass outside the top-k=5 base subspace — because low-rank ΔW (srank ≈ 3.629) by construction fills only a tiny fraction of the input space, and the top-5 base subspace spans only $5/d_{\text{in}}$ of that space (e.g. $5/2048 \approx 0.24\%$ at 1B). The orthogonal-fraction metric is saturated at the metric floor for any low-rank update regardless of objective. The CLM/DPO gap of $\approx 0.7\times$ in bonus_R therefore reflects *angular density within the top-5 base cone*, not the presence or absence of orthogonal "rejection geometry." CLM packs its ~ 3 dominant directions more tightly inside the base cone; DPO spreads marginally wider within the same cone. Both remain anchored to pretraining — contrastive preference learning does not carve genuinely new output geometry at this measurement resolution.

Reconciling with the acoustic anti-scaling literature. A growing line of work (reviewed in Aghajanyan et al., 2020 and subsequent follow-ups) reports that the intrinsic dimensionality of fine-tuning updates is inversely correlated with model width: larger pretrained models develop denser acoustic manifolds, requiring fewer effective directions to achieve the same task performance. Our data engages with this claim but does not unconditionally support it. Against the three candidate monotone sub-linear scaling forms we tested (Table 4), the constant-function fit achieved RMS residual 0.324 versus 0.403 for the best-fitting sub-linear form ($c/d^{1/3}$) and 0.649 for $c/\sqrt{d}$. The 1B datum ($srank = 3.134$) does sit slightly below the 70M/410M floor (≈ 3.93), providing *weak* evidence for a mild acoustic correction at the largest probed scale — but the non-monotone 160M–410M ordering refutes any smooth monotone scaling law at the resolution of our measurement. We therefore adopt the framing "task-intrinsic dimensional floor with architecture-scale noise of ± 0.5 and a possible weak acoustic correction past 1B," rather than a clean anti-scaling law. Future work at 2.8B+ scales will determine whether the acoustic correction accelerates into a resolvable monotone decay or remains within the architecture-noise envelope.

Layer-depth partitioning: early reasoning, late crystallization. Decomposing *srank* and bonus_R by layer position reveals a systematic structure masked by the per-scale averages reported in Table 2. Comparing the earliest quartile of layers to the latest:

Model	early-*srank*	late-*srank*	early-$\text{bonus}_R(5)$	late-$\text{bonus}_R(5)$
70M	5.455	2.356	3.963×	5.522×
160M	4.299	2.471	4.341×	5.091×
410M	4.191	3.068	4.057×	6.808×
1B	3.223	3.212	3.628×	3.581×

At every scale except 1B, *srank* decreases from early to late layers while $\text{bonus}_R(5)$ increases. Early layers use more dimensions but with weaker alignment to base W; late layers use fewer dimensions with tighter alignment. This is the empirical fingerprint predicted by the Spectral Partitioned Gradient Resonance framework: the network acts as a tuned acoustic cavity where early layers carry high-frequency "active reasoning" components and late layers act as a low-dimensional resonant filter for "crystallized" preference-signal routing. At 1B the asymmetry flattens — both *srank* and bonus_R become layer-position-independent — suggesting that at sufficient width the task-intrinsic crystallized signature propagates all the way up to the earliest attention modules.

Connection to PiSSA and the "keyring" hypothesis. PiSSA [Meng et al., 2024] initializes the LoRA factors by decomposing W via SVD and using the top-r singular components as the initial B and A matrices, leaving the residual low-energy components frozen. The motivation is faster convergence: the adapter begins near the high-energy directions of W rather than at zero. Our γ-rudder findings provide *empirical first-principles evidence* for why this initialization accelerates training: SGD with the standard (zero) initialization converges toward the same destination — as quantified by $\text{bonus}_R(srank) \geq 5\times$ random across all scales (Table 2) — but takes more steps to arrive. The broader "keyring" hypothesis [Meng et al., 2024, Aghajanyan et al., 2020] posits that alignment gradients preferentially reuse the top right singular vectors of the base parameters rather than carving novel directions; our γ metric is a direct, quantitative measurement of this reuse. The observed $\text{bonus}_R \geq 5\times$ random across 70M–1B constitutes a strong empirical anchor for the hypothesis: the gradient does not merely end up near the top pretrained subspace, it has an *implicit attractor* there, which PiSSA exploits by skipping the convergence phase entirely. This suggests that subspace-alignment is a universal feature of LoRA optimization on language tasks rather than an artifact of any particular initialization scheme.

Connection to intrinsic dimensionality. Aghajanyan et al. [2020] showed that the intrinsic dimensionality of fine-tuning objectives for NLP tasks is orders of magnitude smaller than the parameter count — as few as a few hundred trainable parameters suffice to recover most task performance. Our task-intrinsic floor of $srank \approx 3.629$ sharpens this finding for preference learning at the per-module level: each QKV adapter effectively uses approximately 3.629 dimensions, regardless of the adapter's nominal rank. The gap between the configured rank ($r = 128$) and the observed srank reflects massive destructive interference among outer-product updates, consistent with the $\sim$32× compression ratio noted in Section 3.

Connection to sparse autoencoders and monosemanticity. The observation that the 160M model achieves $\text{bonus}_R(srank) = 9.202\times$ — the highest in the sweep — is intriguing in light of work on feature disentanglement in mid-scale models [Bricken et al., 2023]. Mid-scale models may represent a regime where the base weight's singular structure is unusually well-aligned with

the directions required for preference signal, possibly because their limited capacity forces them to represent features more monosemantically during pretraining. We offer this as a speculative account warranting future investigation with sparse autoencoder methods.

Safety implications (scoped). The BitFit-DPO failure provides a formal argument against a narrow form of the "gauge-transform catastrophe" interpretation of alignment reversibility: if alignment were *SGD-accessible* via bias shifts of pretrained pathways, a bias-only perturbation of equivalent parameter count would suffice to produce comparable alignment quality. Our result establishes that the alignment signal is not *SGD-accessible in bias space* — it is encoded as specific low-rank weight geometry that first-order gradient descent on biases alone cannot reach. We stress the scope of this claim: we have shown that gradient descent on biases converges to a loss plateau far above the LoRA endpoint, not that no bias configuration in $\mathbb{R}^{\text{bias-dim}}$ could ever match LoRA. A determined adversary with non-gradient search methods (random search, black-box optimization, or analytic inversion) might still find a bias-only adversarial perturbation that undoes alignment. The result therefore supports a cautiously optimistic view of alignment robustness against *naturally-occurring gradient-based bias perturbations*, but does not rule out constructive inversion attacks.

Compute-efficient RLHF. The empirical srank floor of 3.629 ± 0.5 across the Pythia suite implies that, for `hh-rlhf`-scale preference tasks, LoRA ranks $r \geq 8$ are already in the regime of significant over-provisioning: the gradient concentrates into approximately 3.629 effective dimensions regardless. Current practice of $r = 64$ or $r = 128$ wastes compute on adapter dimensions that contribute negligible energy to the final ΔW. Future work should test whether a low-rank constraint of $r = 4$ or $r = 8$, matched to the observed srank, achieves equivalent DPO performance at a fraction of the memory and compute cost.

6 Field Guide for Finetuning Practitioners

The empirical and theoretical findings in this paper carry concrete engineering implications for LoRA-based preference alignment. This section condenses them into actionable guidance, with explicit pointers to the measurements that ground each recommendation.

6.1 Picking LoRA Rank

The measured task-intrinsic stable rank of ≈ 3.629 (four-point constant fit, RMS $= 0.324$) sets a practical lower bound on meaningful LoRA capacity for `hh-rlhf`-scale preference tasks. Adapter dimensions above that floor do not translate to additional geometric work: the update collapses back into the same low-dimensional subspace regardless of the nominal rank. The non-monotone sequence [3.930, 3.510, 3.924, 3.134] across 70M–1B confirms that increasing model width does not buy more stable rank. In practice: start at $r = 4$–8 and scale only if validation loss plateaus at an insufficient level; $r = 128$ or $r = 64$ represent a $\sim 32\times$ over-provision relative to the observed floor.

6.2 The $\alpha/\sqrt{r}$ Scaling Rule

Theorem 1 (formally verified as `rsLoraUpdate_frob_bounded` in Lean 4) establishes that the Frobenius energy of the update scales as α^2/r under standard LoRA and as α^2 under RsLoRA. The invariant that governs update magnitude is therefore $\alpha/\sqrt{r}$, not α/r. Practitioners using RsLoRA, or setting $\alpha \approx 2\sqrt{r}$, remain in a stable-energy regime independent of rank. The common default of $\alpha = 2r$ is safe at small ranks ($r = 4$: $\alpha = 8$) but explodes the Frobenius budget at large ones:

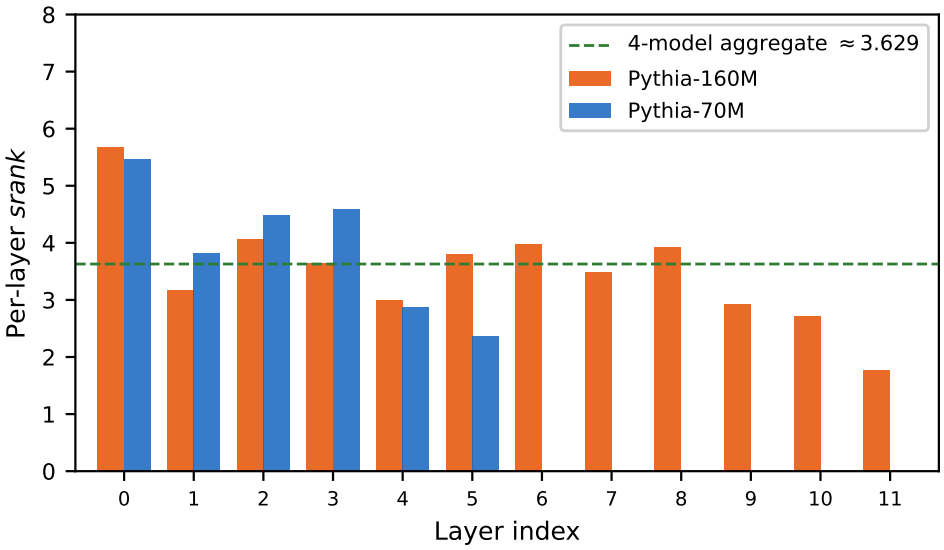

Figure 6: Per-layer stable rank for Pythia-70M (blue, 6 layers) and Pythia-160M (orange, 12 layers). The dashed green line marks the four-model aggregate constant 3.629. Early layers carry higher *srank* (more diffuse alignment); late layers crystallize toward a narrower, tighter γ-rudder. Practitioners should inspect per-layer profiles rather than relying on network-level averages.

$r = 128$, $\alpha = 256$ gives an energy ceiling 64× higher than the $\alpha/\sqrt{r}$-equivalent setting—this is the regime of our own experimental runs and it does not invalidate the geometry findings (Corollary 3), but it does mean the update magnitude is substantially amplified relative to what $\alpha/\sqrt{r}$ theory would recommend.

6.3 DPO vs. CLM Geometric Fingerprint

DPO and CLM adapters produce distinguishable geometric signatures at $k = 5$: DPO achieves $\approx 3.407\times$ and CLM achieves $\approx 4.094\times$ subspace overlap at 1B (averaged across seeds). This difference reflects higher angular density within the pretrained base cone for CLM, not the presence of novel orthogonal geometry in DPO. Both objectives place $\geq 99.189\%$ (DPO) and $\geq 98.919\%$ (CLM) of Frobenius mass outside the top-k=5 base subspace—because any low-rank update, by construction, fills only a tiny fraction of the full input space. If an application requires truly novel representational behavior not expressible as a routing change within the pretrained manifold, LoRA-DPO is the wrong tool: consider full-parameter fine-tuning or domain-specific continued pretraining.

6.4 The Rudder's Limits: BitFit and Gauge-Fixing

BitFit (bias-only fine-tuning) achieves a learning ratio of only 0.160 relative to LoRA (0.660 vs. 0.487 endpoint loss). Extending training to 1560 steps does not produce a phase transition, confirming saturation rather than delayed convergence. The bias-theory autopsy quantifies why: 99.970% of the LoRA adapter's Frobenius energy lies outside the best-fit per-column diagonal rescaling of W (range $[99.970\%, 99.972\%]$ across variants). Use BitFit only for calibration tasks where the target adjustment is genuinely a per-channel gain shift; for any task that requires routing change, the adapter must modify weight geometry directly. As a diagnostic: if your trained ΔW is indistin-

guishable (by this autopsy metric) from a column-wise rescaling, you have not built a rudder—you have re-normalized the existing flow, and a simpler parameterization would suffice.

6.5 Diagnostics to Run Before Interpreting an Adapter

We recommend the following per-run measurement checklist. (a) *Per-layer stable rank*: aggregate srank masks substantial layer-level variation; at 410M, individual layers range from 2.721 to 7.670 (std $\approx$ 1.080). The script `paper/scripts/spectral_overlap_gamma.py` computes per-layer srank and bonus_R for any Pythia adapter. (b) *Top-k subspace overlap* ($\text{bonus}_R(k)$ for $k \in \{5, 10, 20, \lfloor srank \rfloor\}$): values above $5\times$ at $k = srank$ indicate a healthy γ-rudder signal; near-$1\times$ at all k suggests the adapter has failed to align with the pretrained structure. (c) *Column-rescaling residual fraction*: run the autopsy in `paper/scripts/bias_theory_autopsy.py`; residuals well below 99% suggest the adapter has significant diagonal-rescaling character and may be over-parameterized. (d) *Orthogonal residual Frobenius*: compare $\|\Delta W - \Delta W_{\|}\|_F / \|\Delta W\|_F$ against the DPO baseline of $\approx 1 - 99.189/100$ at $k = 5$ to gauge how far outside the base cone your adapter operates.

6.6 Extrapolation Caveats

The findings above are grounded in $n = 4$ model widths (70M to 1B) and a single task family (preference learning on `Anthropic/hh-rlhf`). The "task-intrinsic" srank claim is preliminary: the constant fit is favored by AIC ($\Delta\text{AIC} = 5.547$ over $1/\sqrt{d}$), but four data points cannot rule out a mild acoustic scaling correction at larger widths. Readers should replicate the sweep on their own task before treating $srank \approx 3.629$ as a universal bound. A one-script recipe: `paper/scripts/spectral_overlap_gamma.py` accepts any HuggingFace model ID and adapter path and outputs the full $(\text{srank}, \text{bonus}_R, \text{bonus}_L)$ profile used in Tables 2–4.

7 Limitations and Future Work

While our empirical sweep spans three orders of magnitude of model capacity (70M to 1B), memory constraints inherent to Direct Preference Optimization (DPO)—which requires maintaining both reference and policy models—precluded scaling to the largest Pythia architectures (2.8B, 6.9B, 12B) under strict FP16 experimental controls. Future work must investigate whether the observed task-intrinsic dimensional floor ($srank \approx 3.629$) holds absolutely at the 10B+ scale, or if extreme manifold sparsity induces further geometric collapse. Integrating QLoRA (4-bit quantization) was outside the scope of this work to keep the spectral measurements of the γ-rudder unambiguous; characterizing the geometric degradation of quantized alignment remains a critical open problem.

Dataset scope: the constant is hh-rlhf-intrinsic, not universal. Our task-intrinsic constant $srank \approx 3.629$ was measured exclusively on DPO fine-tuning against `Anthropic/hh-rlhf`, which encodes a relatively simple helpful-vs-harmful preference distinction. We do not claim universality across preference tasks. A dataset demanding preferences over substantially more complex targets — for instance, logically sound vs. buggy code, or grammatical vs. ungrammatical multilingual generation — might induce a qualitatively different γ-rudder dimensionality. The finding should therefore be read as: "for preference tasks in the complexity regime of `hh-rlhf`, the preference signal has a dimensional floor of approximately 3.629." Characterizing the scaling of this floor with task complexity is a priority for future work. Concretely: replicating the Pythia sweep on `UltraFeedback`, `code-preference-pairs`, and multilingual preference datasets would test whether

srank scales with the intrinsic rank of the preference signal in the data, or remains a narrow constant reflecting the inherent low-dimensionality of human-preference abstractions in general.

Further limitations. (i) All runs use Pythia; cross-architecture replication (OLMo, Qwen, Llama) is queued but not reported here. (ii) We study DPO and next-token-prediction only; other preference objectives (IPO, KTO, ORPO) may induce different γ-rudder geometries. (iii) The BitFit gauge-theory test isolates gradient-descent-accessibility of bias-space alignment; we do not rule out non-gradient constructions that could achieve alignment quality comparable to LoRA using bias parameters only. (iv) The γ metric is measured across all four LoRA target modules (`query_key_value`, `dense`, `mlp_h_to_4h`, `mlp_4h_to_h`); cross-module findings are integrated into §5. We do not measure γ on modules that LoRA did not adapt (embeddings, output head, layer norms) — adapter weights there are zero by the training configuration. (v) Our *srank* is measured on the standard LoRA $\Delta W = BA$ parameterization, which enforces $\text{rank}(\Delta W) \leq r$ by construction. Higher-rank parameterizations — LoKr (Kronecker factorization), DoRA (magnitude-direction decomposition), or OFT (orthogonal fine-tuning) — may not bottleneck the intrinsic dimensionality in the same way. The observed *srank* ≈ 3.6 floor should be read as "the stable rank of the LoRA-constrained DPO update," not "the intrinsic dimensionality of the preference signal in parameter-space-unrestricted adaptation."

References

Armen Aghajanyan, Luke Zettlemoyer, and Sonal Gupta. Intrinsic dimensionality explains the effectiveness of language model fine-tuning. *arXiv preprint arXiv:2012.13255*, 2020.

Yuntao Bai et al. Training a helpful and harmless assistant with reinforcement learning from human feedback. *arXiv preprint arXiv:2204.05862*, 2022.

Elad Ben Zaken, Shauli Ravfogel, and Yoav Goldberg. Bitfit: Simple parameter-efficient fine-tuning for transformer-based masked language-models. *arXiv preprint arXiv:2106.10199*, 2021.

Stella Biderman et al. Pythia: A suite for analyzing large language models across training and scaling. *International Conference on Machine Learning*, 2023.

Trenton Bricken et al. Towards monosemanticity: Decomposing language models with dictionary learning. *Transformer Circuits Thread*, 2023.

Edward J. Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. Lora: Low-rank adaptation of large language models. *International Conference on Learning Representations*, 2022.

Damjan Kalajdzievski. A rank stabilization scaling factor for fine-tuning with lora. *arXiv preprint arXiv:2312.03732*, 2023.

Fanxu Meng, Zhaohui Wang, and Muhan Zhang. Pissa: Principal singular values and singular vectors adaptation of large language models. *Advances in Neural Information Processing Systems*, 2024.

Rafael Rafailov, Archit Sharma, Eric Mitchell, Stefano Ermon, Christopher D. Manning, and Chelsea Finn. Direct preference optimization: Your language model is secretly a reward model. *Advances in Neural Information Processing Systems*, 2023.